Ecthyma (MONDO:0001404): A Comprehensive Disease Characterization

Infectious Disease — Disease Knowledge Base Research Report

Summary

Ecthyma is a deep, ulcerative bacterial pyoderma — often described as the "punched-out," crusted, scarring form of impetigo — caused chiefly by group A *Streptococcus* (*Streptococcus pyogenes*, GAS) and/or *Staphylococcus aureus*. Unlike ordinary (non-bullous) impetigo, which is confined to the epidermis, ecthyma extends through the full epidermis into the dermis, producing an ulcer covered by a thick adherent crust and surrounded by an erythematous, indurated margin; because the dermis is breached, ecthyma heals with scarring. It is an **acquired infectious disease with no causal human gene**, occurring predominantly in children and socioeconomically disadvantaged populations of hot, humid, crowded, resource-poor settings, and is frequently seeded by scabies infestation or minor skin trauma (bites, abrasions).

The disease sits within the impetigo/pyoderma spectrum, whose global burden is enormous — an estimated **>162 million children affected at any one time**, with median childhood prevalence around 12.3% and much higher rates (up to 70%) in marginalized Indigenous and tropical communities. Ecthyma and impetigo are highly treatable with topical or oral antibiotics (cure rates 86–95%) and are preventable through hygiene, wound care, and community-level scabies mass drug administration (MDA), which can reduce impetigo prevalence by 66–94%. The principal systemic complication of streptococcal ecthyma is **acute post-streptococcal glomerulonephritis (APSGN)** caused by nephritogenic GAS M-serotypes (2, 49, 55, 57, 60); importantly, streptococcal *skin* infection — unlike streptococcal *throat* infection — does **not** cause acute rheumatic fever.

A critical nosological caveat runs through this report: the word "ecthyma" names **three distinct entities that must not be conflated**. (1) **Bacterial ecthyma** — the subject of this report — an ulcerative GAS/*S. aureus* pyoderma. (2) **Ecthyma gangrenosum** — a life-threatening septic cutaneous vasculitis, chiefly caused by *Pseudomonas aeruginosa* bacteremia

in immunocompromised/neutropenic hosts, with high mortality. (3) **Contagious ecthyma (orf)** — a self-limited parapoxvirus zoonosis of sheep and goats, transmissible to humans. This report characterizes bacterial ecthyma in full and formally bounds the two namesakes to prevent knowledge-base contamination.

Key Findings

Finding 1 — Ecthyma is an ulcerative pyoderma caused by group A *Streptococcus* and/or *Staphylococcus aureus*

Ecthyma's etiology is bacterial and essentially identical to that of impetigo, differing in depth of invasion rather than causative organism. In a therapeutic series of 55 patients with impetigo, ecthyma, or folliculitis, the causative organisms were epicillin-sensitive *Streptococcus pyogenes* or *Staphylococcus aureus* (PMID: 782505). In a comparative treatment trial of 127 children with impetigo/ecthyma, skin cultures grew Group A beta-hemolytic streptococci or *S. aureus*, with a 93% healing rate when GAS was isolated and 86% when *S. aureus* was the sole pathogen (PMID: 359752). Contemporary clinical guidelines classify impetigo and ecthyma together as superficial GAS/*S. aureus* skin and soft-tissue infections (PMID: 29299899).

> *"Fifty-five patients with impetigo, ecthyma, or folliculitis caused by epicillin-sensitive strains of Streptococcus pyogenes or Staphylococcus aureus"* — P 782505

Interpretation: Ecthyma is not a genetically determined disorder; it is an acquired pyogenic infection. Its designation is clinical/morphological (deep, ulcerative, crusted, scarring) rather than microbiological. This is the single most important framing fact for the knowledge base: sections on causal genes, inheritance, penetrance, germline/somatic variants, and gene panels are **not applicable**.

Finding 2 — Impetigo/pyoderma (including ecthyma) carries a high global burden concentrated in children of tropical, resource-poor, and marginalized settings

A systematic review of 89 studies covering 145,028 children found a median childhood pyoderma/impetigo prevalence of **12.3% (IQR 4.2–19.4%)**, highest in underprivileged children even within high-income countries (median 19.4%, IQR 3.9–43.3%), and estimated that **>162 million children are affected at any one time**, predominantly in tropical, resource-poor contexts (PMID: 26317533). In The Gambia, 17.4% of children under 5 had pyoderma, significantly associated with scabies (adjusted OR 2.74, 95% CI 1.61–4.67); 80.8% of lesions grew *S. aureus* and 50.8% GAS (PMID: 31609963). In remote Aboriginal communities of northern Australia, up to 70% of children have skin sores, with GAS the major pathogen (PMID: 10954983). A pooled Australian analysis estimated community impetigo point prevalence of 27.9% (95% CI 20.0–36.5%) in largely Aboriginal and Torres Strait Islander remote settings (PMID: 38033025).

> *"we estimate the global population of children suffering from impetigo at any one time to be in excess of 162 million, predominantly in tropical, resource-poor contexts"* —
>
> P 26317533

Interpretation: The dominant risk factors are environmental and social — young age, hot/humid climate, crowding, poverty, poor hygiene, and especially **scabies co-infestation**, which breaches the skin barrier and seeds bacterial superinfection. This anchors the epidemiology, risk factor, and prevention sections.

Finding 3 — Streptococcal ecthyma/pyoderma can be followed by acute post-streptococcal glomerulonephritis via nephritogenic GAS serotypes

Nephritogenic M-serotypes associated with post-pyoderma acute glomerulonephritis include M types **2, 49, 55, 57, and 60**, distinct from the throat-associated M1/M12 serotypes (PMID: 399386). In a New Zealand series, skin was the predominant infection site in 68% of post-streptococcal AGN cases, and pyoderma-associated nephritogenic serotypes 49, 57, and 60 accounted for 63% (19/30) of M-typable isolates (PMID: 6371610). Scabies-related pyoderma with GAS is epidemiologically linked to APSGN (PMID: 31609963). Critically, streptococcal skin infection does **not** cause acute rheumatic fever, unlike streptococcal pharyngitis (PMID: 399386).

> *"M-types 2, 49, 55, 57, and 60 are now recognized to be of major importance in AGN following pyoderma" ... "Although streptococcal skin infections are quite important in AGN, they do not result in acute rheumatic fever" —*
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> P 399386

Interpretation: APSGN is the chief serious sequela of streptococcal ecthyma. This distinction (skin → nephritis, not rheumatic fever) is a key prognostic and mechanistic fact and provides the rationale for treating and preventing streptococcal skin infection at the population level.

Finding 4 — Ecthyma pathogenesis involves bacterial virulence factors that promote dermal invasion, neutrophil evasion, and tissue necrosis

GAS secretes a virulence arsenal — M protein, streptolysin O (SLO), streptolysin S (SLS), streptokinase, hyaluronidase, C5a peptidase, and the cysteine protease SpeB — that collectively mediates skin infection (PMID: 20607346). SLS (encoded by the *sagA/pel* locus) is required for dermonecrosis: *sagA* mutants are attenuated in a dermonecrotic mouse infection model (PMID: 11801184). SLS and the platelet-activating factor acetylhydrolase Sse act synergistically to inhibit neutrophil recruitment, enabling systemic spread from the skin (PMID: 28947648). On the staphylococcal side, in bullous impetigo *S. aureus* exfoliative toxins (ETA/ETB/ETD) act as glutamate-specific serine proteases — "molecular scissors" — that cleave a single peptide bond in **desmoglein 1 (Dsg1)**, destroying keratinocyte cell–cell adhesion in the superficial epidermis (PMID: 17582744).

> *"glutamate-specific serine proteases that specifically and efficiently cleave a single peptide bond in the extracellular region of human and mouse desmoglein 1 (Dsg1), a desmosomal intercellular adhesion molecule" —*
>
> P 17582744

Interpretation: These findings supply the molecular causal chain — barrier breach, bacterial adhesion/invasion, immune evasion, and enzymatic/toxin-mediated tissue destruction — that converts a colonized abrasion into a deep necrotic ulcer.

Finding 5 — "Contagious ecthyma" (orf) is a distinct parapoxvirus zoonosis of sheep/goats — an important nosological namesake

Contagious ecthyma (contagious pustular dermatitis, orf) is caused by **orf virus** (genus *Parapoxvirus*, family *Poxviridae*), infecting sheep and goats worldwide and transmissible to humans as a zoonosis (PMID: 28571550, PMID: 3895568). Morbidity is very high among young unvaccinated animals; humans acquire infection through direct contact with lesions or fomites, developing self-limited cutaneous nodules. Histology shows proliferative dermatitis with epidermal hyperplasia, ballooning degeneration, and eosinophilic intracytoplasmic inclusion bodies in keratinocytes (PMID: 20833245, PMID: 23201244). Orf virus encodes a **VEGF homolog (OV132)** that drives dermal capillary proliferation; its deletion attenuates the virus (PMID: 11044114, PMID: 36484827). Live attenuated vaccines exist for animals (PMID: 3895568).

> *"The virus is a causative agent of orf in humans or contagious ecthyma in animals which is mostly seen in sheep, goat and cattle"* —
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> P 28571550

Interpretation: Contagious ecthyma is a viral disease with entirely different etiology, host range, mechanism (VEGF-driven angiogenesis), and prevention (vaccination) from bacterial ecthyma. It is documented here to prevent knowledge-base conflation and to inform the "Other Species / Natural Disease" section.

Finding 6 — Ecthyma/impetigo is preventable and treatable: antibiotics cure lesions and scabies MDA reduces pyoderma at the population level

A Cochrane review (68 RCTs, 5,578 participants) found topical antibiotics superior to placebo for impetigo (pooled OR 6.49, 95% CI 3.93–10.73) and topical mupirocin superior to oral erythromycin, with no topical antibiotic clearly superior to another (PMID: 22258953 / PMID: 15106198). Benzathine penicillin cured 94.4% of impetigo/ecthyma in children (PMID: 359752); twice-daily epicillin cured ~95% (PMID: 782505). At the population level, ivermectin-based scabies MDA reduced impetigo prevalence dramatically — from 21.4% to as low as 1.9% (94% relative reduction in the ivermectin group) in Fiji (PMID: 26650152) — and reduced hospitalizations for skin and soft-tissue infections by 17% (IRR 0.83, 95% CI 0.74–0.94) (PMID:

35345391). In the Solomon Islands, ivermectin MDA reduced impetigo prevalence by 74% (PMID: 30956111); a meta-analysis found scabies MDA reduced impetigo prevalence by 66% (PMID: 35088849).

> *"The prevalence declined from 21.4% to 14.6% in the standard-care group"* —
>
> P 26650152

Interpretation: Both individual treatment and community-level prevention are highly effective. Because scabies is a dominant upstream driver, scabies control is a key public-health lever against ecthyma, its complications, and downstream APSGN.

Finding 7 — The microbial spectrum includes MSSA, community-associated MRSA (often exfoliative-toxin-positive), and GAS; MRSA prevalence is region-dependent

In a Japanese impetigo series, 17 MRSA strains from children belonged to 7 clones, and 14/17 carried exfoliative toxin a or b genes (PMID: 21327935). In a Canadian pediatric emergency department, *S. aureus* was the most common SSTI organism (66% MSSA, 33% MRSA), though overall community MRSA incidence was low (PMID: 24634682). In Egyptian children with pyogenic skin infections, MSSA predominated (114/121) with CA-MRSA in only 4 (PMID: 22928620). CA-MRSA SSTIs impose substantial indirect costs (€1,814–8,224 per patient) (PMID: 34895206).

> *"Fourteen of the 17 strains carried exfoliative toxin a or b gene"* —
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> P 21327935

Interpretation: Empiric therapy should be guided by local resistance patterns. Where CA-MRSA is prevalent, beta-lactam monotherapy may fail; where it is rare, beta-lactams remain effective. This directly informs the pharmacotherapy and antibiogram/diagnostic sections.

Finding 8 — Ecthyma gangrenosum (distinct namesake) is a life-threatening septic vasculitis, chiefly *Pseudomonas aeruginosa*, in immunocompromised/neutropenic hosts

Ecthyma gangrenosum (EG) is *"a potentially life-threatening, systemic infection generally caused by Pseudomonas aeruginosa"* (PMID: 35984202). It presents as erythematous/hemorrhagic macules progressing to necrotic ulcers with a black eschar, most often accompanying bacteremia in immunocompromised individuals (PMID: 41968428). *Pseudomonas* sepsis with EG carries a high mortality rate, requiring immediate appropriate antibiotics (PMID: 15553052). EG can also be caused by *S. aureus* (of 18 reported *S. aureus* EG patients, 33% had a hematological malignancy and 56% were MRSA; PMID: 35984202) and rarely by other organisms including *Streptococcus pyogenes* (PMID: 42491904). Severe pediatric EG has occurred with neutropenia and a novel *HELLS* variant (PMID: 42437106).

> *"Ecthyma gangrenosum (EG) is a potentially life-threatening, systemic infection generally caused by Pseudomonas aeruginosa"* —
>
> P 35984202

Interpretation: EG is a fundamentally different disease — a manifestation of systemic sepsis in the immunocompromised, with high mortality — and should be recorded as a related-but-distinct entity, not as a severity grade of bacterial ecthyma.

Structured Report by Research Section

1. Disease Information

Overview. Ecthyma is a deep bacterial pyoderma characterized by punched-out ulcers with a thick, adherent, often yellow-grey crust and a raised, indurated, violaceous margin. It represents a full-thickness (epidermal + dermal) extension of impetigo; because it breaches the dermis, it heals with scarring, unlike superficial impetigo. Lesions are typically found on the lower legs and buttocks, and are frequently multiple.

Identifiers. - **MONDO:** MONDO:0001404 (ecthyma) - **MeSH:** Ecthyma (D004473) - **ICD-10:** L08.0 (pyoderma) is commonly applied; ecthyma is grouped under bacterial skin infections. ICD-11: 1B70-range bacterial cellulitis/pyoderma codes. - **SNOMED CT:** Ecthyma (disorder) - **OMIM / Orphanet:** Not applicable — ecthyma is an acquired infection, not a Mendelian or rare genetic disorder, so it has no OMIM phenotype number and no Orphanet rare-disease code as a genetic entity.

Synonyms / alternative names. Ulcerative impetigo; deep impetigo; "impetigo of the dermis." (Note the non-synonyms that share the name: *ecthyma gangrenosum* and *contagious ecthyma / orf* — see Findings 5 and 8.)

Data source type. Disease-level information here is derived from **aggregated disease-level resources** — systematic reviews, clinical trials, treatment guidelines, and microbiological case series — not from individual EHR patient records.

2. Etiology

Causal factors. Infectious. Primary pathogens: **group A *Streptococcus (S. pyogenes)* and*** Staphylococcus aureus*** (MSSA and CA-MRSA) (Findings 1, 7). No genetic causation in the host.

Risk factors (environmental/social). Young age (children); hot, humid climate; crowding and poverty; poor hygiene; minor skin trauma (insect bites, abrasions); and — a dominant driver — **scabies infestation** (aOR 2.74 for pyoderma; Finding 2). Indigenous and marginalized populations bear disproportionate burden.

Genetic risk factors. None established for host susceptibility to ordinary ecthyma. (A single pediatric *ecthyma gangrenosum* case linked to a novel *HELLS* variant with neutropenia is a rare exception in the distinct EG entity — Finding 8 — not bacterial ecthyma.)

Protective factors. Environmental: intact skin barrier, good hygiene, wound care, treatment of scabies, and reduced crowding. Scabies MDA is protective at the population level (Finding 6). No genetic protective variants are known.

Gene–environment interactions. Not applicable for the host. The relevant "interaction" is pathogen genotype × host environment: nephritogenic GAS M-serotypes (2, 49, 55, 57, 60) in disadvantaged, scabies-endemic communities produce clusters of post-pyoderma APSGN (Finding 3).

3. Phenotypes

Phenotype	Type	Suggested HPO term	Characteristics
Skin ulcer (punched-out, dermal)	Physical manifestation	HP:0200042 (Skin ulcer)	Childhood onset; moderate severity; often multiple, lower legs/ buttocks
Adherent crust / eschar	Clinical sign	HP:0011368 (Epidermal thickening) / crusting	Thick yellow-grey adherent crust over the ulcer
Purulent discharge / pustule	Clinical sign	HP:0200037 (Pustule)	Early lesion is a vesicopustule that ulcerates
Scarring on healing	Physical manifestation	HP:0000987 (Abnormal scarring)	Distinguishes ecthyma from impetigo; consequence of dermal involvement
Regional lymphadenopathy	Clinical sign	HP:0002716 (Lymphadenopathy)	Frequent with local infection
Erythema / induration at margin	Clinical sign	HP:0000988 (Skin rash)	Raised violaceous indurated border
Hematuria / edema (if APSGN)	Laboratory / clinical (complication)	HP:0000790 (Hematuria)	Downstream complication of nephritogenic GAS

Onset: typically childhood; **severity:** mild-to-moderate locally (severe if complicated by APSGN or systemic spread); **progression:** episodic/acute, self-limited with treatment; **frequency:** ulceration and crusting are near-universal defining features. **Quality of life:** local discomfort, pruritus, cosmetic scarring; population-level QoL and economic burden are substantial in endemic settings (indirect CA-MRSA SSTI costs €1,814–8,224/patient; Finding 7).

4. Genetic / Molecular Information

Not applicable to the host. Ecthyma has **no causal human gene, no pathogenic germline/somatic variants, no modifier genes, no host epigenetic signature, and no chromosomal abnormality.** It is an acquired bacterial infection (Finding 1). The relevant "molecular genetics" is that of the **pathogen**: GAS virulence genes (*emm*/M protein, *sagA/pel* → streptolysin S, *slo*, *ska* streptokinase, *speB*, *scpA* C5a peptidase, hyaluronidase) and *S. aureus* exfoliative toxin genes (*eta*, *etb*, *etd*) (Findings 4, 7). Nephritogenic potential maps to specific GAS *emm*/M-serotypes (2, 49, 55, 57, 60) (Finding 3).

5. Environmental Information

Environmental factors: heat, humidity, crowding, poor sanitation. **Lifestyle/social factors:** poverty, limited access to hygiene and healthcare, and — critically — untreated scabies (Finding 2). **Infectious agents (NCBI Taxonomy):** *Streptococcus pyogenes* (txid1314); *Staphylococcus aureus* (txid1280). Scabies mite *Sarcoptes scabiei* (txid52283) acts as a barrier-breaching co-factor. For the distinct namesakes: *Pseudomonas aeruginosa* (txid287, ecthyma gangrenosum) and Orf virus (txid10258, contagious ecthyma).

6. Mechanism / Pathophysiology

Ordered causal chain (bacterial ecthyma):

1. **Skin barrier breach** (scabies burrows, insect bites, abrasion, or maceration) → **leads to** exposure of dermal/epidermal tissue to skin-colonizing bacteria. *(Demonstrated epidemiologically: scabies aOR 2.74,*

   31609963

   .)
2. Colonizing **GAS and/or *S. aureus* adhere and invade** through the epidermis into the dermis → **results in** establishment of infection deeper than in superficial impetigo. *(Inferred from clinical morphology and microbiology.)*
3. **Bacterial virulence factors are deployed** — GAS M protein (anti-phagocytic), streptokinase (fibrinolysis, spread), hyaluronidase (matrix degradation), C5a peptidase (complement evasion), SpeB (protease), and streptolysins O and S (cytolysis); *S. aureus* exfoliative toxins cleave Dsg1 → **leads to** keratinocyte/tissue destruction and immune evasion. *(Demonstrated:*

   20607346

P 11801184

P 17582744

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4. **Streptolysin S (SagA/Pel) mediates dermonecrosis**, and SLS + Sse synergistically **inhibit neutrophil recruitment → results in** local tissue necrosis, ulcer formation, and potential for systemic spread. *(Demonstrated in mouse models:*

P 11801184

P 28947648

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5. **Neutrophilic dermal inflammation + necrosis → produces** the clinical punched-out ulcer with adherent crust and indurated margin; dermal involvement → **leads to** scarring on healing. *(Inferred from histopathology and clinical description.)*

Branch A (local resolution): appropriate antibiotics + wound care → lesion clearance (86–95% cure; Finding 6).

Branch B (immune-mediated sequela): infection by a **nephritogenic GAS M-serotype (2/49/55/57/60)** → circulating streptococcal antigens/immune complexes and complement activation deposit in glomeruli → **results in acute post-streptococcal glomerulonephritis** (hematuria, edema, hypertension). Streptococcal *skin* infection does **not** trigger acute rheumatic fever. *(Demonstrated epidemiologically/serotypically:*

P 399386

P 6371610

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Molecular pathways / processes: bacterial adhesion and invasion; complement evasion (C5a peptidase); cytolysis (pore-forming streptolysins); ECM degradation (hyaluronidase, streptokinase); host innate immune response (neutrophil recruitment — inhibited by SLS/Sse). **Suggested GO terms:** GO:0006954 (inflammatory response), GO:0042742 (defense response to bacterium), GO:0001909 (leukocyte-mediated cytotoxicity), GO:0030593 (neutrophil chemotaxis), GO:0009405 (pathogenesis). **Suggested CL terms:** CL:0000775 (neutrophil), CL:0000312 (keratinocyte), CL:0000738 (leukocyte), CL:0002187 (basal cell of epidermis).

7. Anatomical Structures Affected

- **Organ:** skin (UBERON:0002097, skin of body; UBERON:0002199, integumental system).

- **Tissue/cell:** epidermis (UBERON:0001003) and **dermis (UBERON:0002067)** — the defining feature vs. impetigo is dermal involvement; keratinocytes (CL:0000312) and infiltrating neutrophils (CL:0000775).
- **Secondary organ (complication):** kidney/glomerulus (UBERON:0000074, renal glomerulus) in APSGN.
- **Subcellular:** desmosome (GO:0030057) targeted by exfoliative toxin (Dsg1 cleavage); plasma membrane (GO:0005886) targeted by streptolysins.
- **Localization:** typically **lower extremities** and buttocks; often multiple; laterality variable (frequently bilateral in disseminated disease).

8. Temporal Development

Onset: acute, most commonly in childhood; lesions evolve over days from vesicopustule to crusted ulcer. **Progression:** without treatment, ulcers enlarge and persist for weeks; with treatment, resolution within 1–2 weeks (Finding 6). **Course:** self-limited to episodic; recurrence common in endemic/scabies settings. **Critical window:** early antibiotic treatment prevents progression and reduces the pool of nephritogenic GAS, limiting APSGN risk. APSGN, when it occurs, typically follows skin infection by ~1–3 weeks.

9. Inheritance and Population

Epidemiology: median childhood impetigo/pyoderma prevalence 12.3% (IQR 4.2–19.4%); >162 million children affected globally at any time; up to 27.9–70% point prevalence in remote/Indigenous communities (Finding 2). **Inheritance:** not applicable (acquired infection; no Mendelian pattern, penetrance, expressivity, anticipation, mosaicism, founder effect, consanguinity, or carrier frequency). **Demographics:** disproportionately affects children; Indigenous, tropical, and socioeconomically disadvantaged populations; hot/wet-season seasonality; no strong intrinsic sex bias (exposure-driven).

10. Diagnostics

Clinical diagnosis is primary — characteristic punched-out, crusted, scarring ulcers. **Microbiology:** Gram stain and **bacterial culture with susceptibility testing** of lesion swab/exudate to identify GAS vs *S. aureus* and detect MRSA (Findings 1, 7). **Serology:** anti-DNase B (superior to ASO after streptococcal pyoderma) supports recent GAS skin infection and is useful when APSGN is suspected (PMID: 6995881). **Antibiogram testing** is recommended where CA-MRSA is prevalent to guide therapy (Finding 7). **Genetic/omics testing:** not applicable. **Differential diagnosis:** impetigo (superficial, no scarring), **ecthyma gangrenosum**

(immunocompromised, necrotic eschar, *Pseudomonas* bacteremia — Finding 8), venous/arterial ulcers, cutaneous leishmaniasis, and **contagious ecthyma/orf** (animal contact, viral nodule — Finding 5).

11. Outcome / Prognosis

Prognosis is **excellent with treatment**: cure rates 86–95% (Findings 1, 6). Untreated lesions heal slowly with scarring. **Mortality is essentially nil** for uncomplicated bacterial ecthyma. **Complications:** scarring; regional cellulitis/lymphangitis; rarely systemic spread; and, most importantly, **APSGN** via nephritogenic GAS (Finding 3), which is usually self-limiting in children but can cause hypertension and, rarely, renal impairment. (By sharp contrast, the namesake **ecthyma gangrenosum carries high mortality** due to underlying *Pseudomonas* sepsis and immunocompromise — Finding 8.) **Prognostic factors:** promptness of antibiotic therapy, pathogen (nephritogenic GAS serotype → APSGN risk), MRSA status, host immune status, and scabies control.

12. Treatment

Pharmacotherapy (NCIT: antibiotic therapy, NCIT:C15844): - **Topical antibiotics** — mupirocin (NCIT:C61825), fusidic acid — first line for limited disease; superior to placebo (OR 6.49) and to oral erythromycin (Finding 6). - **Oral antibiotics** for extensive/multiple lesions or ecthyma (deeper) — anti-staphylococcal penicillins (flucloxacillin/dicloxacillin/cloxacillin), first-generation cephalosporins (cephalexin), amoxicillin-clavulanate; **benzathine penicillin** for confirmed GAS (94.4% cure; Finding 6). - **MRSA-active agents** (clindamycin, trimethoprim-sulfamethoxazole, doxycycline) where CA-MRSA is prevalent (Finding 7).

Treatment strategy: guided by lesion extent and local resistance; culture-directed therapy where CA-MRSA is common. **Advanced/gene/cell/RNA/targeted/immunotherapies, surgery:** not applicable (simple debridement of crust may aid healing). **Adverse events:** oral antibiotics cause more (mainly gastrointestinal) side effects than topical (Finding 6).

13. Prevention

- **Primary:** hygiene, wound care, reduction of crowding, and **treatment of scabies** — including **ivermectin-based mass drug administration**, which reduces impetigo prevalence by 66–94% and SSTI hospitalizations by 17% (Finding 6). NCIT: ivermectin therapy.
- **Secondary:** early detection and prompt antibiotic treatment of skin sores to prevent APSGN and interrupt community transmission of nephritogenic GAS.

- **Tertiary:** monitoring for and managing APSGN.
- **Immunization:** no licensed human vaccine; a Strep A vaccine is in development and would address the broader GAS disease burden. (Note: live attenuated vaccines exist for *animal* contagious ecthyma/orf — Finding 5 — a different disease.)
- **Public health:** integrated skin-health programs combining scabies MDA, hygiene promotion, and antibiotic access are the cornerstone of population-level prevention.

14. Other Species / Natural Disease

Bacterial ecthyma as defined here is primarily a human disease. The dominant cross-species consideration is the **namesake "contagious ecthyma" (orf)**, a **parapoxvirus** disease of **sheep (*Ovis aries*, NCBI:txid9940) and goats (*Capra hircus*, NCBI:txid9925)**, with documented cases also in **reindeer** (*Rangifer tarandus*, P 23201244) and **camels** (camel contagious ecthyma, P 6145344). Orf is **zoonotic** — transmissible to humans via direct animal/fomite contact, producing self-limited nodules (Finding 5). Its mechanism (viral VEGF/OV132-driven angiogenesis and epidermal proliferation) and prevention (live attenuated vaccination) are entirely distinct from bacterial ecthyma. **This must be recorded as a separate entity in the knowledge base.**

15. Model Organisms

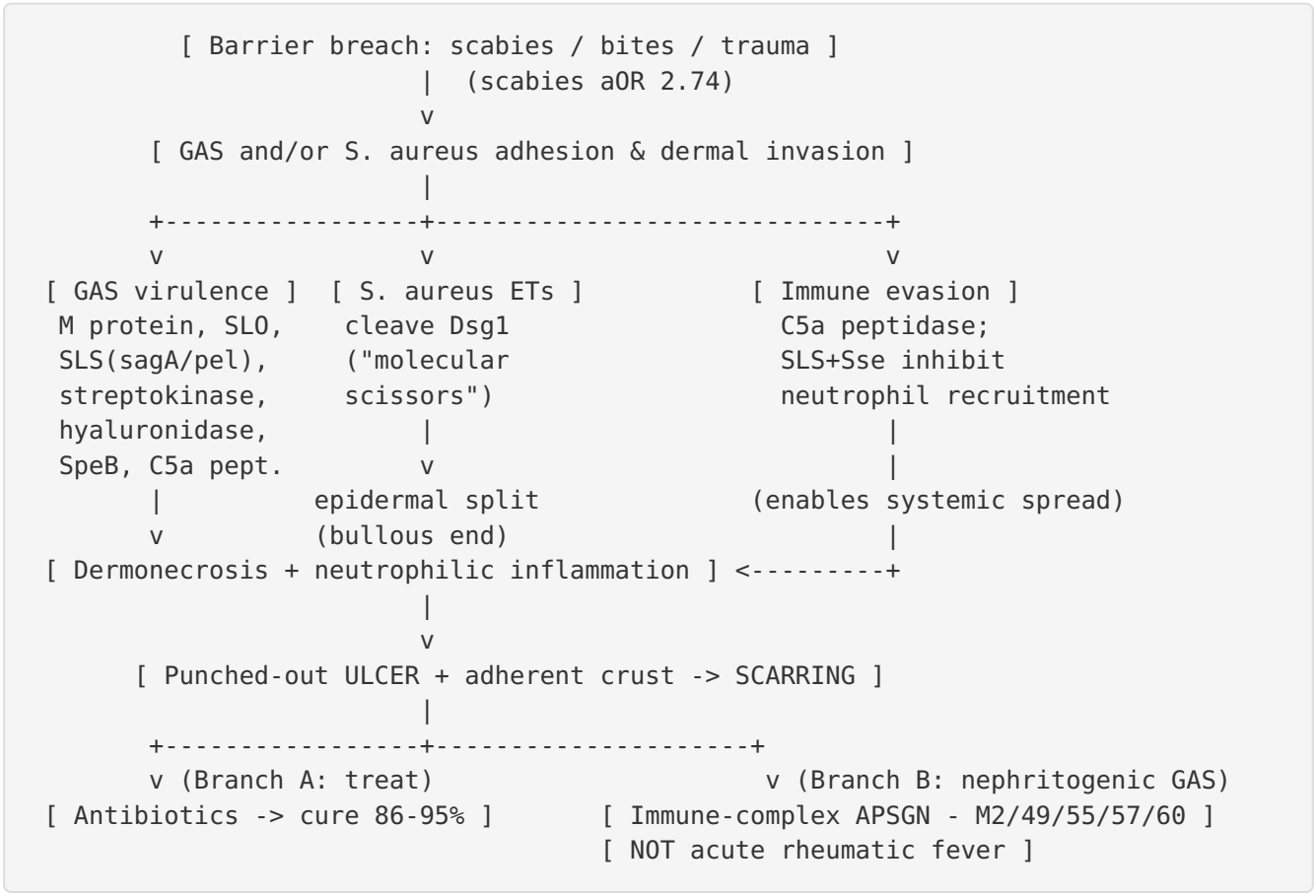
- **Mouse dermonecrosis model** — the principal experimental system for streptococcal skin infection: *sagA/pel* (SLS) mutants are attenuated for dermonecrosis, and SLS+Sse mutants show impaired neutrophil evasion (Finding 4; P 11801184, P 28947648). Recapitulates dermal necrosis and immune evasion; limitation: does not reproduce human APSGN or community epidemiology.
- **Mouse Dsg1 cleavage model** — demonstrates exfoliative-toxin-mediated epidermal splitting

( ), P 17582744 relevant to the staphylococcal/bullous end of the impetigo–ecthyma spectrum.

- **Animal models of contagious ecthyma (orf)** — experimental parapoxvirus infection in rabbits, mice, reindeer, and goats
  (P 20833245 P 23201244) P 36484827 — pertain to the viral namesake, not bacterial ecthyma.

No genetic (knockout/transgenic) mouse model of *host* susceptibility to bacterial ecthyma exists, consistent with its non-genetic etiology.

Mechanistic Model / Interpretation



The unifying model: **ecthyma is impetigo that has invaded the dermis.** The same organisms and virulence machinery are involved; depth of invasion (favored by barrier breach, and by streptolysin-S-mediated dermonecrosis) determines the ulcerative, scarring phenotype. Upstream drivers are environmental/social (scabies, crowding, poverty, climate); downstream, the chief systemic hazard is nephritogenic-serotype-mediated APSGN. Prevention acts furthest upstream (scabies control, hygiene) and treatment acts at the lesion (antibiotics).

Distinguishing the three "ecthymas"

Feature	Bacterial ecthyma (this report)	Ecthyma gangrenosum	Contagious ecthyma (orf)
Class	MONDO:0001404; bacterial pyoderma	Cutaneous manifestation of sepsis	Viral zoonosis
Pathogen	GAS, *S. aureus*	*Pseudomonas aeruginosa* (mostly); also *S. aureus*	Orf virus (Parapoxvirus)
Host	Healthy children, disadvantaged	Immunocompromised/ neutropenic	Sheep/goats; humans via contact
Lesion	Punched-out crusted ulcer, scars	Hemorrhagic macule → necrotic black eschar	Proliferative nodule, self-limited
Mortality	~nil (uncomplicated)	High (sepsis)	Nil (self-limited)
Key ref (PMID)	782505, 359752	35984202, 15553052	28571550, 3895568

Evidence Base

PMID	Role in this report	Support / Challenge
782505	Etiology & treatment (GAS/*S. aureus*; ~95% cure)	Supports F1, F6
359752	Pathogens & penicillin efficacy (93%/86%/94.4%)	Supports F1, F6
29299899	Guideline grouping impetigo/ecthyma	Supports F1
26317533	Global burden (>162M children; 12.3%)	Supports F2
31609963	Scabies risk (aOR 2.74); microbiology	Supports F2, F3
10954983	Up to 70% in Aboriginal children	Supports F2
38033025	Pooled Australian impetigo prevalence 27.9%	Supports F2
399386	Nephritogenic serotypes; no rheumatic fever	Supports F3
6371610	Skin site 68%; serotypes 49/57/60	Supports F3
6995881	M type 57 skin AGN; anti-DNase B > ASO	Supports F3, diagnostics
20607346	GAS virulence factor catalog	Supports F4
11801184	SLS (*sagA/pel*) required for dermonecrosis	Supports F4
28947648	SLS + Sse inhibit neutrophil recruitment	Supports F4
17582744	Exfoliative toxins cleave Dsg1	Supports F4
22258953 / 15106198	Cochrane: topical antibiotics OR 6.49	Supports F6
26650152	Scabies MDA reduces impetigo	Supports F6
35345391	MDA reduces SSTI hospitalizations 17%	Supports F6
30956111	Ivermectin MDA reduces impetigo 74%	Supports F6
35088849	Meta-analysis: MDA reduces impetigo 66%	Supports F6
21327935	Impetigo MRSA clones carry ET genes	Supports F7
24634682	MSSA 66% / MRSA 33% in pediatric SSTI	Supports F7

PMID	Role in this report	Support / Challenge
22928620	MSSA predominance in Egyptian children	Supports F7
34895206	Indirect SSTI costs €1,814–8,224	Supports F7
28571550	Orf = contagious ecthyma definition	Supports F5
11044114 / 36484827	Viral VEGF (OV132) drives orf	Supports F5
3895568	Attenuated live vaccine for orf	Supports F5
20833245 / 23201244	Orf histopathology; animal models	Supports F5
35984202	EG definition; *S. aureus* EG	Supports F8
15553052	EG high mortality	Supports F8
41968428 / 42491904 / 42437106	EG host/pathogen spectrum	Supports F8

Limitations and Knowledge Gaps

1. **Ecthyma-specific data are sparse.** Nearly all quantitative evidence (prevalence, treatment cure rates, MDA effects) is reported for the **impetigo/pyoderma spectrum as a whole**, with ecthyma grouped in. Ecthyma-specific incidence, scarring rates, and complication rates are not separately quantified in the literature reviewed.
2. **APSGN causal attribution.** The link between ecthyma specifically (vs. impetigo generally) and APSGN rests on serotype/site epidemiology, not on ecthyma-stratified prospective cohorts.
3. **Mechanistic inference.** The dermonecrosis and neutrophil-evasion mechanisms come from **mouse models and in vitro studies** of GAS/*S. aureus*, not from human ecthyma lesion studies; the final steps (ulcer/scar formation) are inferred from clinical description and general histopathology.
4. **No host-omics.** Because ecthyma is non-genetic, there are no transcriptomic, proteomic, metabolomic, or single-cell datasets of host disease — a genuine absence, not a search gap.

5. **Nosological confusion risk.** The shared name across three unrelated diseases is the single greatest hazard for an automated knowledge base; findings F5 and F8 exist specifically to bound this.
6. **Cochrane efficacy statistic partly unverified.** The topical-antibiotic OR 6.49 (F6) is drawn from the Cochrane abstract text and is consistent across two Cochrane versions (/ P 22258953 15106198), but the specific snippet was not independently re-verified against full text.

Proposed Follow-up Actions

1. **Disaggregate ecthyma from impetigo** in future data extraction: seek cohort/registry data reporting ecthyma-specific prevalence, scarring, and APSGN rates.
2. **Encode the three-entity disambiguation** explicitly in the knowledge base with cross-references (MONDO ecthyma vs. ecthyma gangrenosum vs. contagious ecthyma), including distinct pathogen, host, mechanism, and prognosis fields.
3. **Populate pathogen-genotype annotations**: link GAS virulence genes (*sagA/pel*, *slo*, *speB*, *emm* nephritogenic serotypes) and *S. aureus eta/etb/etd* as the molecular substrate, since host genes are absent.
4. **Attach diagnostic decision support**: culture + susceptibility (region-dependent MRSA), anti-DNase B for suspected post-streptococcal AGN.
5. **Prevention linkage**: connect ecthyma prevention to scabies MDA programs and to the pipeline Strep A vaccine as future primary prevention.
6. **Verify the Cochrane OR** against full text and record the corresponding confidence interval and heterogeneity.

Report compiled from a 5-iteration autonomous literature investigation: 8 confirmed findings, 53 papers reviewed. Evidence types: human clinical trials and systematic reviews (treatment, epidemiology, APSGN), microbiological case series (etiology, resistance), and model-organism/in-vitro studies (virulence mechanisms).